

Prince Savsaviya

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EDUCATION

University of California, Riverside | *Master of Science in Computer Science, GPA: 3.59/4.0*

March 2026

Adani Institute of Infrastructure Engineering | *B.E. Information & Communication Technology, GPA: 8.71/10*

June 2024

EXPERIENCE

Author & Maintainer, TethrLink | [Site](#) | [GitHub](#)

Mar 2026 – Present

Open-Source / Python, GStreamer, PipeWire, Wayland/D-Bus, GTK4, Jetpack Compose, MediaCodec

- Launched a Linux-to-Android second-monitor system in Python using GStreamer and PipeWire, streaming at up to 120 FPS at <50ms latency over USB; implemented dual-codec support (JPEG + H.264) with hot-reload FPS and quality tuning
- Architected Wayland-native screen capture via Mutter ScreenCast D-Bus API, creating a true virtual display with configurable resolution (720p/1080p/1440p) and spatial arrangement relative to the primary monitor
- Delivered a GTK4 + Libadwaita desktop GUI with live connection stats, 3-tab settings panel, and pystray system tray; Android client achieves zero-GC rendering via Jetpack Compose + MediaCodec with SurfaceView bitmap recycling and ~3s auto-discovery
- Shipped to Google Play Store (Closed Alpha track) by configuring Kotlin DSL signing, R8 minification, and ProGuard rules, clearing Play Console review at targetSdk 35 compliance

Graduate Student Researcher

May 2025 – Mar 2026

University of California, Riverside

- Engineered a modular fluid-structure interaction (FSI) simulation pipeline coupling Navier-Stokes fluid, nonlinear solid mechanics, and ALE mesh-motion solvers using FEniCSx, PETSc, and MPI
- Validated 9 standalone benchmarks (CFD1-3, CSM1-3) achieving <0.5% error vs. published references on refined meshes across 6 refinement levels (L0-L5)
- Integrated strong Dirichlet-Neumann coupling with Aitken relaxation and adaptive time stepping, reproducing stable FSI1/FSI2 benchmark oscillatory behaviour
- Built a configuration-driven reproducibility framework supporting 6 mesh refinement levels and checkpoint-based restart, reducing re-run setup from manual scripting to a single config file

Research Intern

May 2023 – Dec 2023

Adani Institute of Infrastructure Engineering

- Devised a hybrid PDE-ML model for Li-Ion battery degradation prediction, reducing MSE by 98% (10.36→0.21), RMSE by 86% (3.22→0.46), and MAE by 65% (0.74→0.26) by integrating physics-informed PDEs with Gaussian Process Regression
- Forecasted State of Charge, State of Health, and Remaining Useful Life of NCA batteries under varying temperature and C-rate conditions using an ensemble of algorithms (GPR, Random Forest, SVM, XGBoost, Neural Networks)
- Co-authored findings in *Journal of Power Sources* (Impact Factor ~8.1) | [Paper](#)

PROJECTS

Wildfire-Twin | *Python, Kafka, Docker, Spark, Apache Sedona, Streamlit, DuckDB, H3* | [GitHub](#)

Jan 2026 – Mar 2026

- Constructed a high-throughput Kafka-to-Spark geospatial pipeline processing live fire streams against 11.5 million building footprints, maintaining a sub-30-second processing SLA during bursts of 1,000 concurrent events
- Reduced P95 processing latency by 91% (215s down to 18.65s) by replacing inefficient Python UDFs with native Sedona Java expressions and executing broadcast spatial joins
- Stabilized Streamlit dashboard rendering to prevent browser crashes at statewide zoom levels using Uber's H3 (Resolution 7) indexing and a low-latency DuckDB analytical sink

Application Load Balancer | *Go, TypeScript, Docker, net/http*

Apr 2025 – Jun 2025

- Programmed a Layer-7 load balancer in Go using net/http, implementing Round Robin, Weighted Round Robin, and Least-Connections routing for concurrency-safe traffic management
- Built a full-stack control plane with Go and TypeScript for real-time configuration of traffic weights and circuit-breaker thresholds
- Deployed a cooldown window mechanism for unstable servers before reintegration to prevent cascading outages; containerized the system with Docker for scalable deployment

RetriVeX | *Python, Django, Elasticsearch, FAISS, BERT, PyLucene* | [GitHub](#)

Jan 2025 – Mar 2025

- Formulated a hybrid semantic and keyword search engine over a 150K+ record drug database, improving query relevance by combining dense vector embeddings (BERT/FAISS) with sparse retrieval (Elasticsearch BM25)
- Achieved sub-second semantic similarity search by generating context-aware PyTorch embeddings and constructing an optimized FAISS vector index
- Shipped a Django REST application with 3 plug-and-play search backends (Lucene, Elasticsearch, BERT/FAISS) for real-time drug information lookup

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, Java, Kotlin, C++, SQL

Backend & Systems: Django, net/http, GStreamer, PipeWire, Wayland/D-Bus, Android SDK, Jetpack Compose, MediaCodec, MPI, FEniCSx, PETSc, Linux

Data & Infrastructure: Docker, Apache Kafka, Apache Spark (Sedona), Elasticsearch, DuckDB, Git

Machine Learning & AI: PyTorch, Scikit-learn, Transformers/BERT (HuggingFace), FAISS, Deep Learning, NLP, Physics-Informed ML, Model Evaluation